

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester II Examination, 2023-24

Paper No. BAS-201: Algebra, Probability and Statistical Methods

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Roll NO. ccscseesseeessesseneees BAS-201 : Algebra, Probability and Statistical Methods

Question 1.

- (a) Define the probability of an event. State and prove the addition law of probability for n events. [#1]
- (b) A card is drawn at random from a deck of ordinary playing cards. What is the probability that it is a diamond, a face card, or a king? [#1]

Question 2.

- (a) What do you understand by conditional probability? State and prove Bayes' theorem. [#1]
- (b) If two fair dice are thrown, what is the probability of getting:
- (i) a double six,
 - (ii) a sum of 8 or more dots.

[#1]**Question 3.**

- (a) Explain the concept of a random variable. Differentiate between discrete and continuous random variables. What is a distribution function and what are its properties? [#1]
- (b) Given the following probability density function:

$$f(x) = A(4x - 2x^2), \quad 0 < x < 2$$

and 0 otherwise. Calculate the value of A so that $f(x)$ is a probability density function. [#1]

Question 4.

- (a) Explain the concepts of Probability Mass Function (p.m.f.) and Probability Density Function (p.d.f.). Define $E(X)$, the expected value of a random variable X . [#1]
- (b) Find the probability distribution of the sum of the dots when two fair dice are thrown. Use the probability distribution to find the probabilities of obtaining:
- (i) a sum of 8 or 11,
 - (ii) a sum that is greater than 8,
 - (iii) a sum that is greater than 5 but less than or equal to 10.

[#1]

Question 5.

- (a) Define the Binomial distribution and derive its mean and variance. [#1]
- (b) A fair coin is tossed 5 times. Find the probabilities of obtaining various numbers of heads. [#1]

Question 6.

- (a) Define Poisson distribution and derive its mean and variance. [#1]
- (b) Write down the steps involved in fitting a Poisson distribution to a given frequency data. [#1]

Question 7.

- (a) Define Normal distribution and derive its mean and variance. [#1]
- (b) What do you mean by Point Estimation? Explain the desirable properties of a good point estimator. [#1]

Question 8. Explain any **four** of the following with examples:

- (a) Null Hypothesis and Alternative Hypothesis
- (b) Simple Hypothesis and Composite Hypothesis
- (c) Acceptance Region and Rejection Region
- (d) Type-I Error and Type-II Error
- (e) One-tailed test and two-tailed test

[#1]

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